

DATA ANALYTICS PROJECT

# MoMo Mobile Topup Performance & Cashback Strategy Analysis

End-to-End Project Documentation

ANALYSIS PERIOD

Jan–Dec 2020

TECHNOLOGY

SQL Server · Power BI

BUSINESS DOMAIN

Fintech · Mobile Topup

**Growth creates value only when incremental contribution offsets incremental cashback cost.**

<https://app.powerbi.com/view?r=eyJrIjoiZmViZTA5YmEtZDI4OC00ZjNlLWlwNzEtN2RmZDBjOWUyNjkyIiwidCI6IjM3MGZiM2I4LTMzMDYtNDg5MCM05MDYzLWNjMDhiZTc4ODI1NyIsImMiOiJFwfQ%3D%3D>

## Executive Summary

MoMo's Mobile Topup service provides broad ecosystem reach and recurring transaction potential, but the proposed increase in cashback rates creates a material trade-off between transaction growth and contribution margin. This project combines SQL Server data preparation with a five-page Power BI report to evaluate historical performance, merchant economics, customer behaviour and cashback sustainability.

**DECISION** Do not proceed with a blanket cashback rollout. Use merchant-specific, capped experiments and scale only when measured incremental contribution is positive.

## Headline Results

| Business area       | Headline result                                                                                     | Decision implication                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Overall performance | 696.60M VND GMV; 18.75M VND gross revenue; 11.79M VND current contribution.                         | Manage performance using contribution as well as volume.                           |
| Growth pattern      | September peaked at 65.4M VND GMV and the highest AOV of approximately 55K VND.                     | Investigate the mix behind the peak and test repeatable drivers.                   |
| Purchase behaviour  | Purchase-for-others represented 16.6% of orders but 36.4% of GMV; AOV was about 2.9× self-use.      | Develop gifting and family top-up propositions.                                    |
| Merchant economics  | Viettel generated 51.6% of GMV but only 38.4% of revenue; Mobifone led contribution at 3.84M VND.   | Separate scale from monetisation and use merchant-specific terms.                  |
| Customer health     | New-user rate fell from 10.3% in March to 6.2% in December; frequency remained near 1.0.            | Separate acquisition from activation and repeat-use tactics.                       |
| Cashback proposal   | Contribution falls 79.5% at 0% uplift; portfolio break-even requires approximately 389% GMV growth. | Reject blanket rollout; validate causality and economics through controlled tests. |

## Document Structure

- Business context, objectives and stakeholder requirements
- Data scope, analytical approach and metric definitions
- Page-by-page report documentation and validated findings
- Integrated recommendations, experimentation framework and success criteria
- Assumptions, limitations and proposed next steps

# 1. Business Context and Objectives

## 1.1 Business Context

MoMo's Mobile Topup service enables users to recharge their own mobile balance or purchase mobile cards for others. The service plays an important role in MoMo's ecosystem due to its broad user reach, recurring transaction potential and contribution to platform revenue.

However, the Vietnamese e-wallet market is highly competitive, with competitors using aggressive cashback programmes to attract and retain users. MoMo is therefore considering increasing cashback rates across all telecommunications merchants.

Before implementing this strategy, MoMo needs to understand the service's current performance, identify the key drivers of revenue and user activity, and evaluate whether the expected growth could offset the additional cashback cost. Using historical transaction data from January to December 2020, customer demographic information and merchant commission rates, this project evaluates Topup performance, user behaviour, merchant contribution and the financial implications of the proposed cashback strategy.

## 1.2 Overall Business Objective

**OBJECTIVE** Develop an end-to-end analytical solution using SQL Server and Power BI that monitors Mobile Topup performance, identifies key growth drivers across time, merchants and customer segments, and evaluates the financial sustainability of MoMo's proposed cashback strategy.

## 1.3 Specific Objectives

1. Measure monthly Mobile Topup performance through transaction, customer and financial metrics.
2. Identify the main drivers behind changes in GMV and commission revenue.
3. Understand customer demographics and transaction behaviour, particularly differences between new and current users.
4. Evaluate merchant-level performance and revenue contribution.
5. Quantify the financial impact of proposed cashback rates and estimate the growth required to offset additional cashback cost.
6. Provide actionable recommendations for Marketing and Product teams while protecting contribution margin.

## 1.4 Core Business Questions

- How is Mobile Topup performing over time?
- Which periods, merchants and customer groups drive GMV and gross revenue?
- Are acquisition and repeat usage improving?
- Does merchant scale translate into monetisation and contribution?
- Can additional GMV realistically offset the proposed cashback cost?

## 2. Business and Reporting Requirements

### 2.1 Stakeholders and Decisions Supported

| Stakeholder            | Decision supported                                                      | Primary measures                                               |
|------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|
| Business / Product     | Assess service health, product usage and sustainable growth priorities. | GMV, orders, users, AOV, frequency, contribution               |
| Marketing              | Design acquisition, activation and targeted cashback campaigns.         | New-user rate, segment value, incremental GMV and contribution |
| Merchant / Partnership | Evaluate merchant economics and negotiate commercially viable terms.    | GMV share, commission rate, revenue share, contribution        |
| Finance                | Protect margin and approve promotional investment.                      | Cashback cost, contribution gap, break-even uplift             |
| Management             | Decide whether and how to scale the cashback proposal.                  | Scenario contribution, risk status, test outcome               |

### 2.2 Functional Requirements

- Monitor orders, active users, GMV, gross revenue, current contribution, AOV, ARPU and purchase frequency.
- Support filtering by Month Year and Merchant across relevant report pages.
- Analyse monthly trends and compare current-period performance with the prior month.
- Compare self-use and purchase-for-others behaviour.
- Rank merchants by GMV, revenue and contribution and compare GMV share with revenue share.
- Analyse new versus current users, demographics, geography, value and RFM segmentation.
- Simulate assumed GMV uplift while keeping proposed cashback rates fixed.
- Calculate contribution gap, required GMV lift and break-even feasibility by merchant.
- Provide dynamic decision guidance and merchant-level scenario detail.

### 2.3 Non-Functional Requirements

- Metric consistency: all totals and rates must use governed measures rather than raw column aggregation.
- Usability: consistent sidebar navigation, filters, titles, colours and tooltip definitions.
- Performance: a star-schema model with one-to-many relationships from dimensions to facts.
- Interpretability: gross revenue, cashback cost and contribution must remain clearly distinguished.
- Decision integrity: scenario assumptions must not be presented as causal outcomes without an experiment.

## 3. Data Scope and Analytical Method

### 3.1 Data Scope

| Data domain     | Key content                                                 | Role in analysis                                  |
|-----------------|-------------------------------------------------------------|---------------------------------------------------|
| Transactions    | Transaction date, amount, user, merchant and purchase type. | Orders, GMV, AOV, activity and trend analysis     |
| Customers       | User identifier, age, gender, location and customer status. | Demographic, acquisition and behavioural analysis |
| Merchants       | Telecommunications merchant and commission rate.            | Revenue, contribution and merchant comparison     |
| Date            | Calendar month and reporting attributes.                    | Time intelligence and month-over-month comparison |
| Scenario inputs | Current cashback, proposed cashback and assumed GMV lift.   | Financial simulation and break-even analysis      |

### 3.2 Analytical Workflow

- Extract and profile source data in SQL Server.
- Clean dates, customer attributes, merchant names and transaction values.
- Create reporting views and reusable business calculations.
- Build a Power BI star schema using date, customer and merchant dimensions.
- Create governed DAX measures for KPIs, segmentation and scenario analysis.
- Validate totals at portfolio, month and merchant levels.
- Translate analytical results into page-level decisions and recommendations.

### 3.3 Core Metric Definitions

| Metric                | Business definition                                                              |
|-----------------------|----------------------------------------------------------------------------------|
| GMV                   | Total value of completed Mobile Topup transactions.                              |
| Gross Revenue         | Commission revenue earned from merchants before cashback cost.                   |
| Current Contribution  | Gross revenue less current cashback cost.                                        |
| Scenario Contribution | Contribution under proposed cashback rates and the selected assumed GMV lift.    |
| Contribution Gap      | Scenario contribution minus current contribution.                                |
| AOV                   | GMV divided by number of orders.                                                 |
| Gross ARPU            | Gross revenue divided by active users.                                           |
| Purchase Frequency    | Orders divided by active users.                                                  |
| New-user Rate         | New users divided by total active users in the reporting period.                 |
| Required GMV Lift     | Current contribution divided by proposed contribution at current GMV, minus one. |

**IMPORTANT** Required GMV lift is a financial threshold, not a forecast. Assumed uplift is a scenario input, not an observed causal effect.

## 4. Report Architecture

The Power BI solution follows a five-page decision journey, moving from context to performance diagnosis and finally to strategy evaluation.

| Page                 | Purpose                                                                       | Primary decision                                                         |
|----------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Introduction         | Explain the problem, objectives, headline insight and recommended direction.  | Why should stakeholders use this report?                                 |
| Business Overview    | Assess service scale, financial performance and purchase behaviour over time. | Where is performance improving or weakening?                             |
| Merchant Performance | Separate merchant scale from monetisation and contribution.                   | Which merchant strategies should be prioritised?                         |
| Customer Insight     | Understand acquisition, demographics, geography and customer value.           | Which customer groups and behaviours should Marketing target?            |
| Cashback Scenario    | Test financial sustainability under different GMV uplift assumptions.         | Should the proposed cashback policy be launched, redesigned or rejected? |

### 4.1 Navigation and Interaction

- A persistent sidebar provides navigation between all analytical pages.
- Month Year and Merchant slicers define the main reporting context.
- Dynamic titles and scenario cards make the active assumption explicit.
- Scenario visuals respond to the assumed uplift where logically appropriate.
- Required GMV lift remains fixed because it is the calculated break-even threshold.
- The sensitivity curve remains fixed while the selected scenario marker moves.

## 5. Page 1 — Business Overview

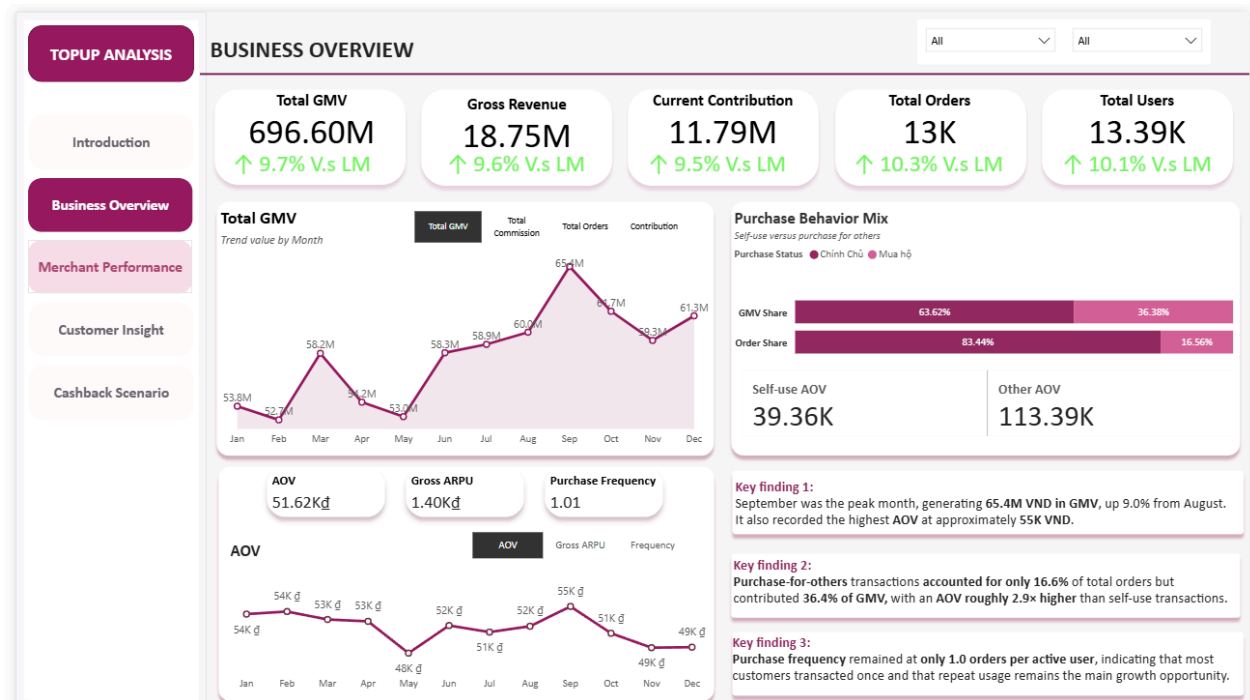


Figure 1. Business Overview dashboard

### 5.1 Page Requirement

Provide an executive view of Mobile Topup scale, financial performance and purchase behaviour, with sufficient time context to identify growth peaks, weak periods and behavioural drivers.

### 5.2 Visual Components

- KPI cards: Total GMV, Gross Revenue, Current Contribution, Total Orders and Total Users with month-over-month change.
- Monthly trend selector for GMV, commission, orders and contribution.
- Purchase Behaviour Mix comparing self-use and purchase-for-others.
- AOV, Gross ARPU and Purchase Frequency trend selector.
- Embedded key-finding narratives for management interpretation.



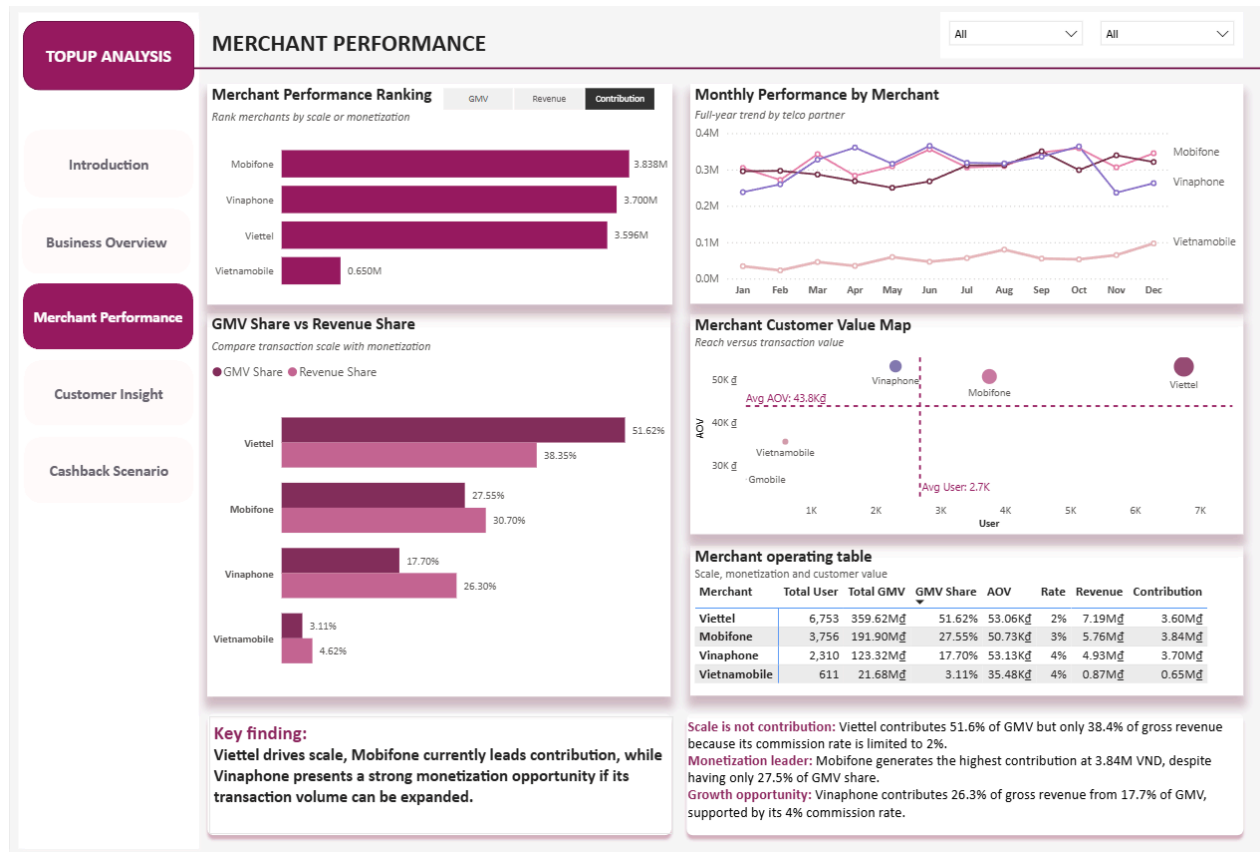
## 5.3 Validated Findings

| Finding                                        | Evidence                                                                        | Business meaning                                                               |
|------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| September was the peak month.                  | 65.4M VND GMV, up 9.0% from August; AOV approximately 55K VND.                  | The peak combines stronger scale and transaction value.                        |
| Purchase-for-others is a high-value behaviour. | 16.6% of orders generated 36.4% of GMV; AOV 113.39K versus 39.36K for self-use. | Low-volume gifting behaviour creates disproportionate transaction value.       |
| Repeat usage is weak.                          | Purchase frequency remained approximately 1.01 order per active user.           | Most users transact once; activation and repeat usage are the main growth gap. |
| Current scale is material.                     | 696.60M VND GMV, 18.75M VND gross revenue and 11.79M VND current contribution.  | Future promotion should protect the existing contribution base.                |

## 5.4 Actionable Recommendations

- Investigate September’s merchant mix, purchase type, active-user growth and AOV to identify repeatable drivers.
- Develop family, gifting or “top up for others” propositions because this behaviour produces substantially higher AOV.
- Prioritise activation and second-purchase journeys rather than relying only on new-user acquisition.
- Track contribution alongside GMV and gross revenue in monthly business reviews.

## 6. Page 2 — Merchant Performance



## 6.3 Merchant Economics

| Merchant     | GMV / share      | Commission | Revenue | Contribution | Interpretation                                    |
|--------------|------------------|------------|---------|--------------|---------------------------------------------------|
| Viettel      | 359.62M / 51.62% | 2%         | 7.19M   | 3.60M        | Scale leader but monetisation is constrained.     |
| Mobifone     | 191.90M / 27.55% | 3%         | 5.76M   | 3.84M        | Contribution leader with balanced scale and rate. |
| Vinaphone    | 123.32M / 17.70% | 4%         | 4.93M   | 3.70M        | Strong monetisation and expansion potential.      |
| Vietnamobile | 21.68M / 3.11%   | 4%         | 0.87M   | 0.65M        | Small scale; selective growth opportunity.        |

## 6.4 Validated Findings

- Scale is not contribution: Viettel generated 51.6% of GMV but only 38.4% of gross revenue because its commission rate is limited to 2%.
- Mobifone generated the highest current contribution at approximately 3.84M VND despite holding 27.5% of GMV.
- Vinaphone generated 26.3% of gross revenue from 17.7% of GMV, supported by its 4% commission rate.
- Viettel has the largest customer base and the highest merchant AOV, making it strategically important but financially sensitive.

## 6.5 Actionable Recommendations

- Use merchant-specific promotional economics rather than a uniform cashback policy.
- Protect Mobifone's contribution leadership and validate whether additional volume can be acquired efficiently.
- Prioritise Vinaphone for measured expansion because its revenue share exceeds its GMV share.
- Renegotiate commission support or reduce cashback intensity for Viettel before any large-scale campaign.
- Evaluate merchants using contribution and incremental contribution, not GMV ranking alone.

## 7. Page 3 — Customer Insight

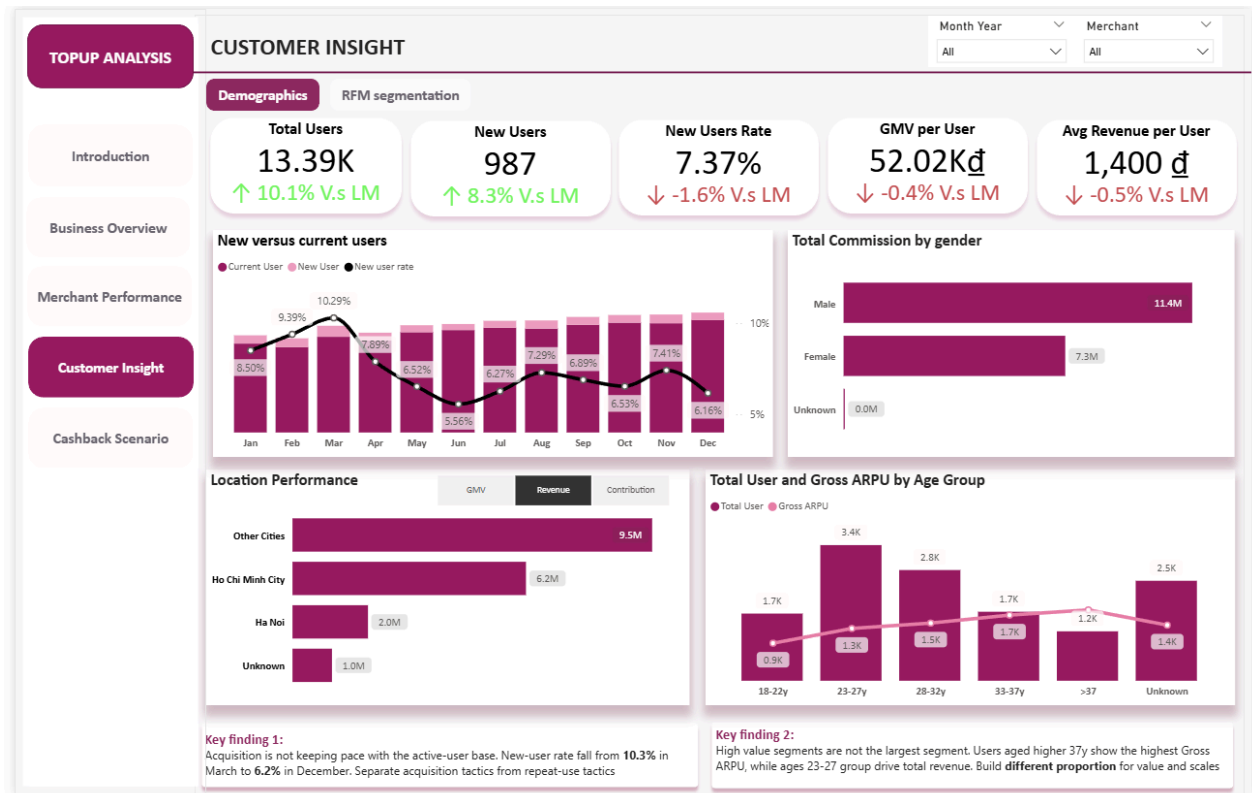


Figure 3. Customer Insight — Demographics dashboard

### 7.1 Page Requirement

Identify acquisition health, customer value differences and geographic or demographic opportunities, while separating strategies for new and current users. The page also provides a navigation path to RFM segmentation for lifecycle targeting.

### 7.2 Visual Components

- Customer KPI cards: Total Users, New Users, New-user Rate, GMV per User and Average Revenue per User.
- Monthly comparison of new and current users with the new-user rate.
- Commission by gender.
- Location performance switcher for GMV, revenue and contribution.
- Total users and Gross ARPU by age group.
- Demographics and RFM segmentation navigation tabs.

## 7.3 Validated Findings

| Finding                                    | Evidence                                                                                                                      | Implication                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Acquisition is weakening.                  | New-user rate fell from 10.29% in March to 6.16% in December; December had 76 new users versus March's 113.                   | Active-user growth is increasingly dependent on current users.            |
| High-value and high-scale segments differ. | Age >37 recorded the highest Gross ARPU, while ages 23–27 generated the most revenue and had the largest known-age user base. | Use distinct value and scale propositions.                                |
| Geographic value is concentrated.          | Other Cities generated about 9.5M VND revenue, ahead of Ho Chi Minh City at 6.2M and Ha Noi at 2.0M.                          | Growth is not limited to the two largest cities.                          |
| Gender contribution is uneven.             | Male customers generated approximately 11.4M VND commission versus 7.3M for female customers.                                 | Diagnose reach, usage and transaction-value differences before targeting. |

## 7.4 Actionable Recommendations

- Separate acquisition, first-transaction activation and repeat-use campaigns; each solves a different funnel problem.
- Target ages 23–27 with frequency and retention mechanics because the group drives scale.
- Develop higher-value, convenience-led propositions for customers older than 37.
- Investigate Other Cities by province and merchant before reallocating location-based marketing budget.
- Use RFM segments to prioritise valuable and at-risk users, and measure incremental contribution by segment.

## 8. Page 4 — Cashback Scenario

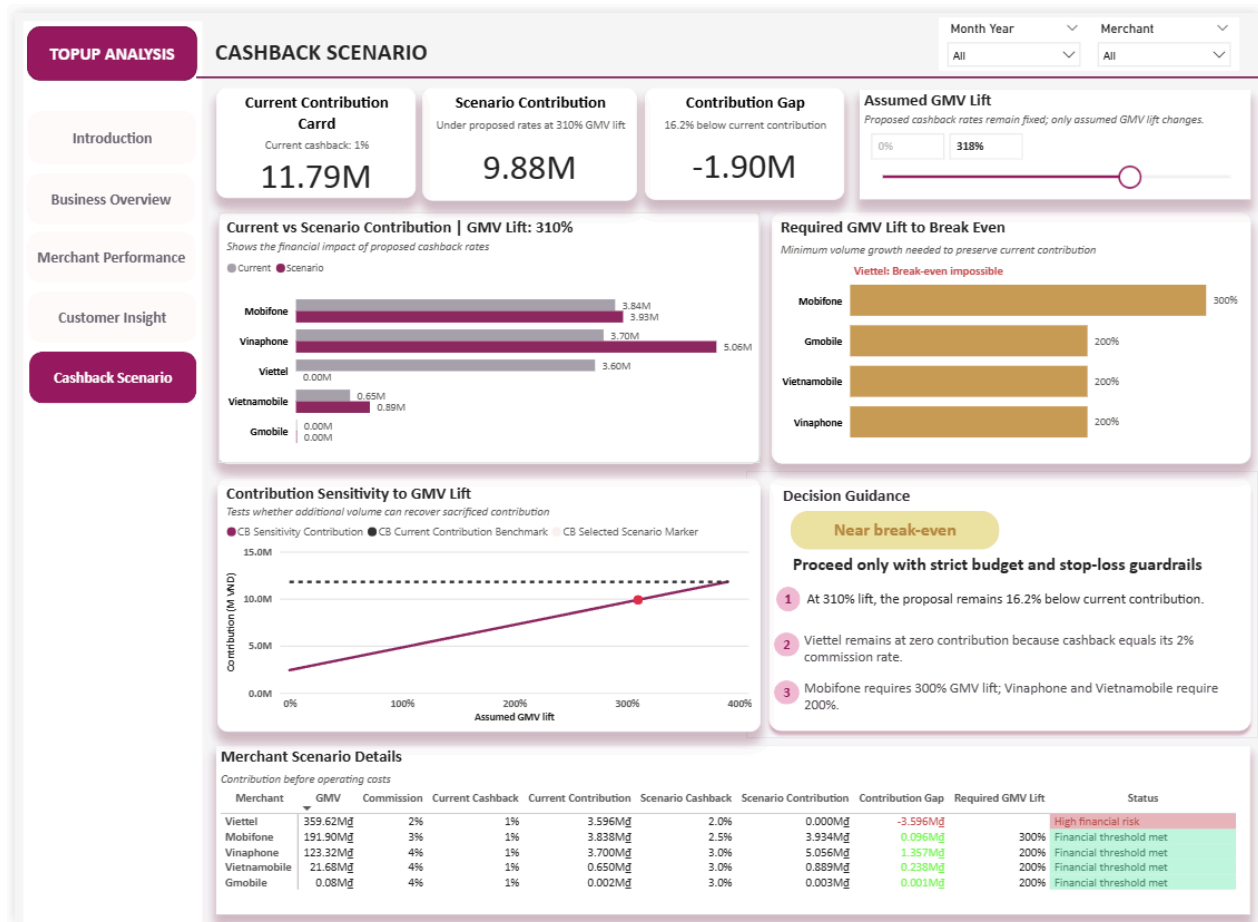


Figure 4. Cashback Scenario dashboard at 310% assumed GMV lift

### 8.1 Page Requirement

Evaluate whether additional transaction volume could offset the margin sacrificed under proposed merchant-specific cashback rates, and translate scenario results into an explicit rollout decision.

### 8.2 Visual Components

- Current Contribution, Scenario Contribution and Contribution Gap cards.
- Assumed GMV Lift scenario control with fixed proposed cashback rates.
- Current versus Scenario Contribution by merchant.
- Required GMV Lift to Break Even by merchant, with impossible break-even warning.
- Contribution sensitivity curve and selected scenario marker.
- Dynamic Decision Guidance and Merchant Scenario Details matrix.

## 8.3 Baseline Financial Impact

| Metric                             | Result             | Interpretation                                             |
|------------------------------------|--------------------|------------------------------------------------------------|
| Current contribution               | 11.79M VND         | Existing contribution that the proposal must protect.      |
| Proposed contribution at 0% uplift | 2.41M VND          | Contribution after proposed rates without incremental GMV. |
| Contribution gap                   | -9.38M VND         | Immediate financial sacrifice before incremental growth.   |
| Contribution reduction             | 79.5%              | The majority of current contribution is lost at baseline.  |
| Portfolio break-even uplift        | Approximately 389% | GMV must rise to about 4.89× the baseline level.           |

## 8.4 Merchant Break-even Findings

| Merchant     | Proposed cashback | Required GMV lift | Decision meaning                                                              |
|--------------|-------------------|-------------------|-------------------------------------------------------------------------------|
| Viettel      | 2.0%              | Impossible        | Cashback equals the 2% commission rate; contribution remains zero at any GMV. |
| Mobifone     | 2.5%              | 300%              | GMV must increase from 100 to 400 to preserve contribution.                   |
| Vinaphone    | 3.0%              | 200%              | GMV must triple to preserve contribution.                                     |
| Vietnamobile | 3.0%              | 200%              | GMV must triple to preserve contribution.                                     |

**AT 310% UPLIFT** Scenario contribution reaches approximately 9.88M VND but remains 16.2% below current contribution. The portfolio is near break-even, not financially recovered.

## 8.5 Actionable Recommendations

- Reject a blanket rollout because moderate uplift does not recover sacrificed contribution.
- Exclude or redesign Viettel's proposed offer; no finite GMV uplift can create contribution at a zero post-cashback margin.
- Run merchant-level, capped A/B tests with an untreated control group.
- Use incremental contribution as the primary decision KPI and establish budget and stop-loss guardrails.
- Scale only when observed incremental contribution is positive and repeat behaviour persists after the incentive ends.

## 9. Integrated Actionable Recommendations

### 9.1 Strategic Recommendation

**RECOMMENDATION** Replace the proposed blanket cashback policy with a merchant-specific, segment-aware and experiment-led growth strategy.

### 9.2 Prioritised Action Plan

| Priority | Action                                                                                 | Owner                 | Success measure                                   |
|----------|----------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------|
| P0       | Redesign Viettel's cashback rate below its commission rate or secure merchant funding. | Partnership + Finance | Positive post-cashback contribution rate          |
| P0       | Define campaign budget caps, stop-loss rules and test approval criteria.               | Finance + Product     | Maximum downside fixed before launch              |
| P1       | Run merchant-level A/B tests rather than a portfolio-wide launch.                      | Product + Data        | Statistically credible incremental contribution   |
| P1       | Build second-purchase and repeat-use journeys for current users.                       | CRM + Product         | Higher 30-day repeat rate and frequency           |
| P1       | Test gifting/family propositions for purchase-for-others behaviour.                    | Marketing             | Incremental GMV and contribution per exposed user |
| P2       | Develop distinct propositions for ages 23–27 and >37.                                  | Marketing             | Segment-level incremental contribution            |
| P2       | Investigate and replicate September's performance drivers.                             | Business + Data       | Validated driver with repeatable uplift           |

### 9.3 Merchant Strategy

- Viettel — Protect economics first. Its scale magnifies the impact of an unsustainable offer.
- Mobifone — Defend contribution leadership and test whether volume can be expanded at an acceptable marginal cost.
- Vinaphone — Prioritise controlled growth because monetisation efficiency is strong relative to GMV share.
- Vietnamobile — Use small, targeted tests; its low scale limits total impact but also limits testing risk.

### 9.4 Customer Strategy

- Acquisition: diagnose the declining new-user rate before increasing incentive spend.
- Activation: move new users to a second transaction quickly through time-bound, capped journeys.
- Retention and value: target frequency and habitual utility, with differentiated propositions for high-scale and high-ARPU segments.

## 10. Cashback Experiment and Measurement Framework

### 10.1 Recommended Test Design

- Randomly assign eligible users to control and treatment groups within each merchant.
- Keep eligibility, campaign window and communication consistent except for cashback exposure.
- Cap cashback per user and in total to limit financial downside.
- Measure outcomes during the campaign and through a post-campaign retention window.
- Evaluate results separately by merchant, new/current status and priority customer segment.



## 10.2 Measurement Hierarchy

| Level      | Metric                              | Purpose                                                          |
|------------|-------------------------------------|------------------------------------------------------------------|
| Primary    | Incremental contribution            | Determines whether the campaign creates financial value.         |
| Secondary  | Incremental GMV and orders          | Measures volume response after removing control-group behaviour. |
| Secondary  | Repeat purchase / frequency         | Tests whether the incentive creates sustained usage.             |
| Guardrail  | Cashback cost and contribution loss | Limits downside during the experiment.                           |
| Diagnostic | Conversion, AOV and active users    | Explains how the observed effect was produced.                   |

## 10.3 Decision Rules

- Scale: incremental contribution is positive, the confidence interval is acceptable and repeat behaviour persists.
- Iterate: GMV increases but contribution remains slightly negative; reduce rate, narrow eligibility or seek merchant funding.
- Stop: incremental cashback cost exceeds incremental gross revenue or a defined loss limit is reached.
- Reject: the post-cashback contribution rate is zero or negative, as in the proposed Viettel case.

**PRIMARY FORMULA**  $\text{Incremental Contribution} = \text{Incremental Gross Revenue} - \text{Incremental Cashback Cost}$

## 10.4 Recommended Monitoring Cadence

| Cadence           | Review focus                                                        | Decision                             |
|-------------------|---------------------------------------------------------------------|--------------------------------------|
| Daily during test | Spend, data quality, loss guardrails and severe merchant anomalies. | Continue or stop immediately.        |
| Weekly            | Incremental GMV, conversion, AOV and contribution by merchant.      | Maintain, narrow or adjust exposure. |
| Post-test         | Full incremental contribution and subgroup results.                 | Scale, iterate or reject.            |
| 30 days after     | Repeat usage and contribution without promotional support.          | Confirm sustainability.              |

## 11. Assumptions and Limitations

- The analysis uses historical data from January to December 2020 and may not reflect current market conditions, customer behaviour or merchant contracts.
- The scenario assumes proposed cashback rates remain fixed and GMV changes proportionally; it does not model capacity constraints, customer heterogeneity or competitive response.
- Required GMV lift is a mathematical break-even threshold and should not be interpreted as a forecast.
- Historical trends and segment differences are descriptive. They do not establish causal campaign impact.
- Unknown demographic and location values may reduce the precision of segment conclusions.
- Commission rates are treated as given. Merchant-funded promotion or renegotiated rates would change the economics.
- Gmobile has negligible baseline GMV in the displayed operating data; conclusions for that merchant should be treated cautiously.

## 12. Conclusion

The Mobile Topup business has meaningful scale, but growth quality varies across purchase behaviours, merchants and customer segments. Purchase-for-others transactions provide a clear high-value use case, while low purchase frequency and a weakening new-user rate indicate that acquisition alone will not solve the growth challenge. Merchant economics also differ materially: Viettel drives scale, Mobifone leads contribution and Vinaphone shows strong monetisation efficiency.

The proposed cashback policy is not financially sustainable as a blanket strategy. At baseline, it reduces contribution by 79.5%, and the portfolio requires approximately 389% GMV uplift to recover the current contribution level. Viettel cannot break even under the proposed terms because cashback equals its commission rate.

**FINAL DECISION** MoMo should pursue targeted, merchant-specific experiments with strict budget controls and scale only when incremental contribution—not headline GMV—is positive.